



Team Name : YODHA



TEAM NAME AND MEMBER DETAILS



MEMBERS :

1. Soumyajit Bhandary
2. Sumandeep Sahoo
3. Atul Modi
4. Rajdeep Thakur

THEME :

AI-Powered Trust & Safety Platform :

(Combating Fraud, Abuse, and Misinformation Using LLMs and Graph Intelligence)

TEAM NAME :

YODHA



PROBLEM STATEMENT & SCOPE



The Problem: Trust is Under Attack in Amazon's Marketplace

Challenges in the Amazon Ecosystem:

1. Fake and incentivized reviews erode product authenticity.
2. Counterfeit or hijacked listings hurt customers and sellers.
3. Review bombing & returns fraud manipulate product rankings.
4. Manual moderation is reactive, not real-time.

Scope of Innovation:

We are building a real-time Trust & Safety Platform for Amazon that:

1. Uses LLMs to detect fake or AI-generated reviews.
2. Leverages graph analytics to catch collusion and hijacks.
3. Provides TrustOps Dashboards for Amazon moderators to act instantly.



WORKING BACKWARDS FROM THE CUSTOMER



Working Backwards: Building for Amazon's Trust & Safety Teams

Primary Customer:

🛡️ Amazon Trust & Safety Operations (T&S Ops)

Need: Early detection of high-risk activity (e.g., fake reviews, seller collusion)

Pain Point: Current rule-based systems have high false positives, low agility

End Users Benefiting:

🛒 Customers: Accurate reviews, trustworthy listings

👤 Genuine Sellers: Protected from review manipulation & listing hijacks

🧠 Amazon Data Analysts: Receive clean, enriched signals for decision-making

Why It Matters to Amazon:

Customer trust is core to Amazon's Flywheel

Faster fraud detection = reduced abuse, improved CSAT & retention



SUCCESS METRICS & EXPECTED IMPACT



Key Outcomes for Amazon :

Quantifiable Metrics:

- >85% detection accuracy for fake/incentivized reviews using LLMs
- <2-minute latency for listing hijack alerts
- 40% reduction in fraudulent return abuse (via behavior analytics)
- Launch of Trust Scores for listings to boost moderation priority

Impact to Amazon:

1. Reduces manual investigation backlog
2. Enhances customer buying confidence
3. Decreases seller complaints and support escalations
4. Scales moderation efficiency with intelligent auto-triage



SCALABILITY AND DOMAIN EXPANSION



Scalable Across Amazon's Global Marketplace :

Built for Amazon Scale:

1. Serverless microservices (AWS Lambda, SageMaker, DynamoDB, S3)
2. Modular pipelines (LLM, graph engine, lifecycle tracker)
3. Plug-and-play for other Amazon surfaces: Prime Wardrobe, Amazon Fresh, Kindle Store
4. Multilingual & Regional Readiness:
5. NLP pipeline supports multi-language review detection
6. Handles localized abuse in India, LATAM, Europe marketplaces

Expansion Opportunities:

-  Logistics fraud (return scams, item-switching)
-  Amazon Advertising fraud (fake product boosts)
-  Voice and video reviews (future LLM agent moderation)
-  Blockchain-backed reviewer trust score



SYSTEM ARCHITECTURE



End-to-End Architecture for Real-Time Trust & Safety

Flow Overview:

Review & Listing Ingestion → Ingests reviews/listings via live feed or batch

Preprocessing Layer → Cleans & prepares data for AI processing

LLM-based Detection → Detects fake/AI-written/incentivized content

Graph Engine (Neo4j) → Models relationships between reviewers, sellers, products

Anomaly Detection Layer → Flags hijacks, review rings, return manipulation

TrustOps Dashboard → Real-time moderator interface for triage & alerts

Tech Stack:

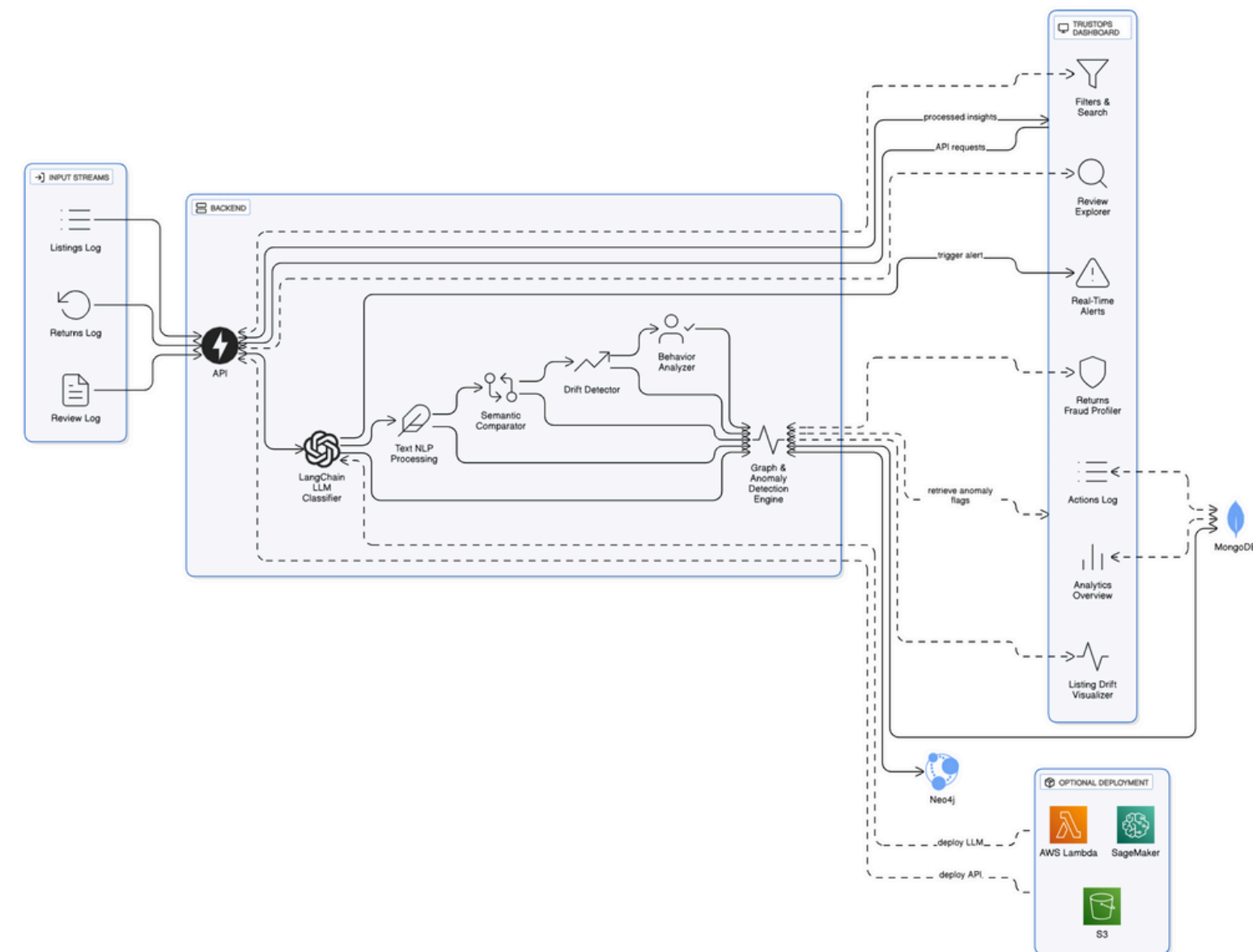
LLMs: OpenAI, LangChain

Graph DB: Neo4j

Backend: Python, FastAPI

Infra: AWS Lambda, SageMaker, S3

Storage: MongoDB, Vector Store





DETECTION LOGIC AND CORE DATA FLOW



How the Detection Pipeline Works :

LLM Review Classification: Flags AI-generated or incentivized reviews

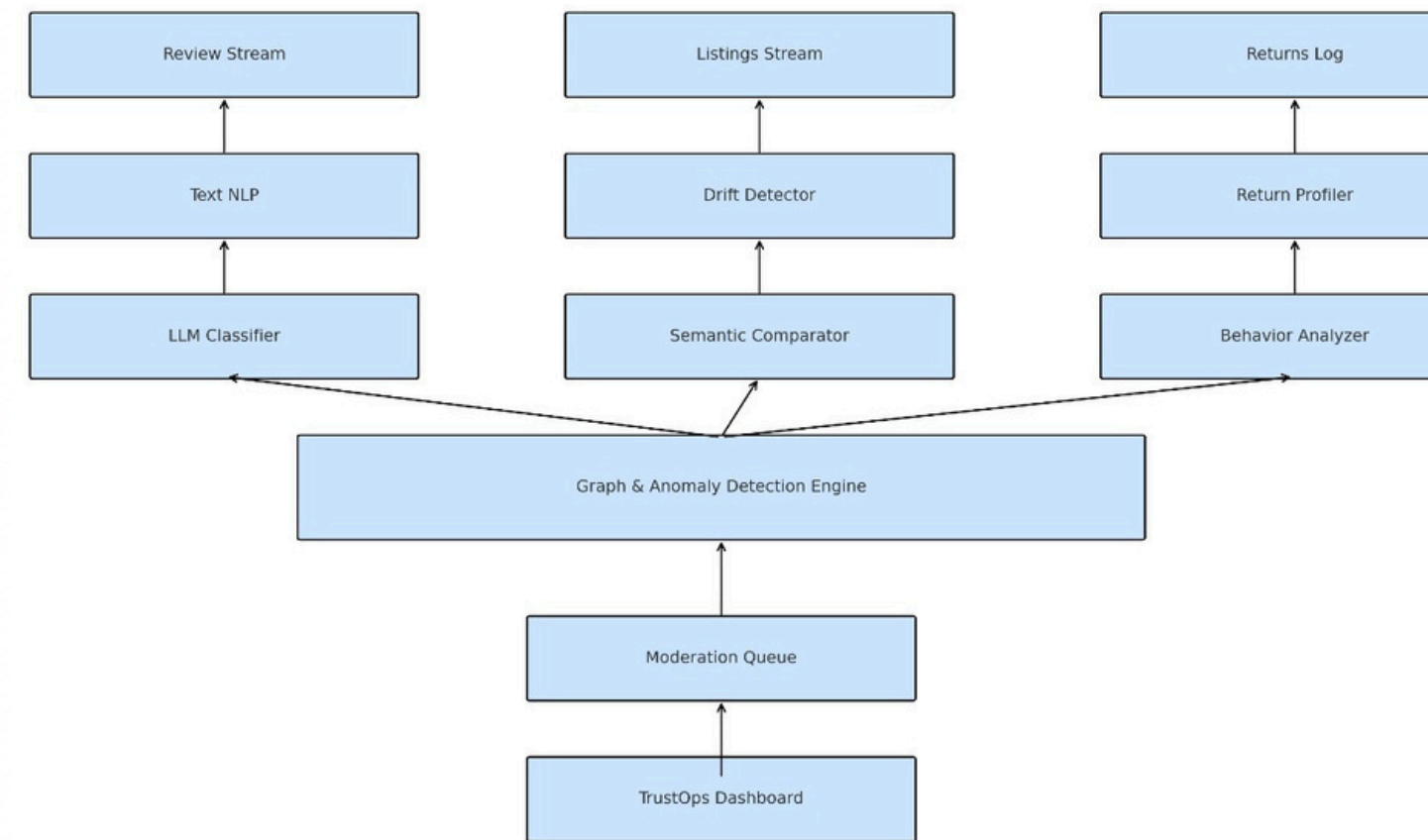
Drift Detection: Monitors title/description mismatches over time
(listing hijacks)

Reviewer-Product Graph Analysis: Detects collusion
(fake networks)

Return Profiling: Identifies unusual return behaviors
(e.g., high-value fraud rings)

Alert Generation:

TrustOps Dashboard receives categorized fraud events in real time.





TRUSTOPS DASHBOARD



Real-Time Alerts Feed :

Displays live fraud alerts generated by the system, including event type, confidence score, and affected listing or review. Moderators can quickly prioritize high-risk incidents for investigation.

TrustOps

Alerts

Reviews

Listings

Returns

Logs

Analytics

Real-Time Alerts

Date Range

Jun 02, 2025 - Jun 09, 2025

Fraud Type

All Types

Search by ID

Review ID, Listing ID, etc.

Status

All Statuses

Apply

Reset

Alerts Feed

Risk	Timestamp	Category	Details	Status	Actions
85%	6/9/2025, 11:00:36 AM	Fake Review	Review ID: R12345	Pending	...
80%	6/9/2025, 10:00:36 AM	Hijacked Listing	Listing ID: L67890	Investigating	...
92%	6/9/2025, 9:00:36 AM	Suspicious Return	Return ID: RET001	Pending	...
40%	6/9/2025, 8:00:36 AM	Fake Review	Review ID: R54321	Approved	...
70%	6/9/2025, 7:00:36 AM	Hijacked Listing	Listing ID: L09876	Rejected	...

Moderator Action Log Feed :

Tracks actions taken by moderators in response to alerts — including dismissals, escalations, or flagged content. Enables auditability, accountability, and continuous feedback for improving the detection model.

TrustOps

Alerts

Reviews

Listings

Returns

Logs

Analytics

Moderator Actions Log

Date Range

Jun 02, 2025 - Jun 09, 2025

Fraud Type

All Types

Search by ID

Item/Moderator ID or Name

Apply

Reset

Log Entries

Timestamp	Moderator	Action	Item Type	Item ID	Reason/Details	View
6/9/2025, 11:59:36 AM	Alice (M001)	Approved	Review	R54321	Review deemed genuine after check.	🔍
6/9/2025, 11:53:36 AM	Bob (M002)	Rejected	Listing	L09876	Clear signs of hijacking, reverted to old version.	🔍
6/9/2025, 11:47:36 AM	Alice (M001)	Investigating	Return	RET001	N/A	🔍
6/9/2025, 11:41:36 AM	Charlie (M003)	Flagged	User	U003	Suspicious return pattern.	🔍



DEEPER INSIGHTS: REVIEWS, LISTINGS, RETURNS



Review Explorer :

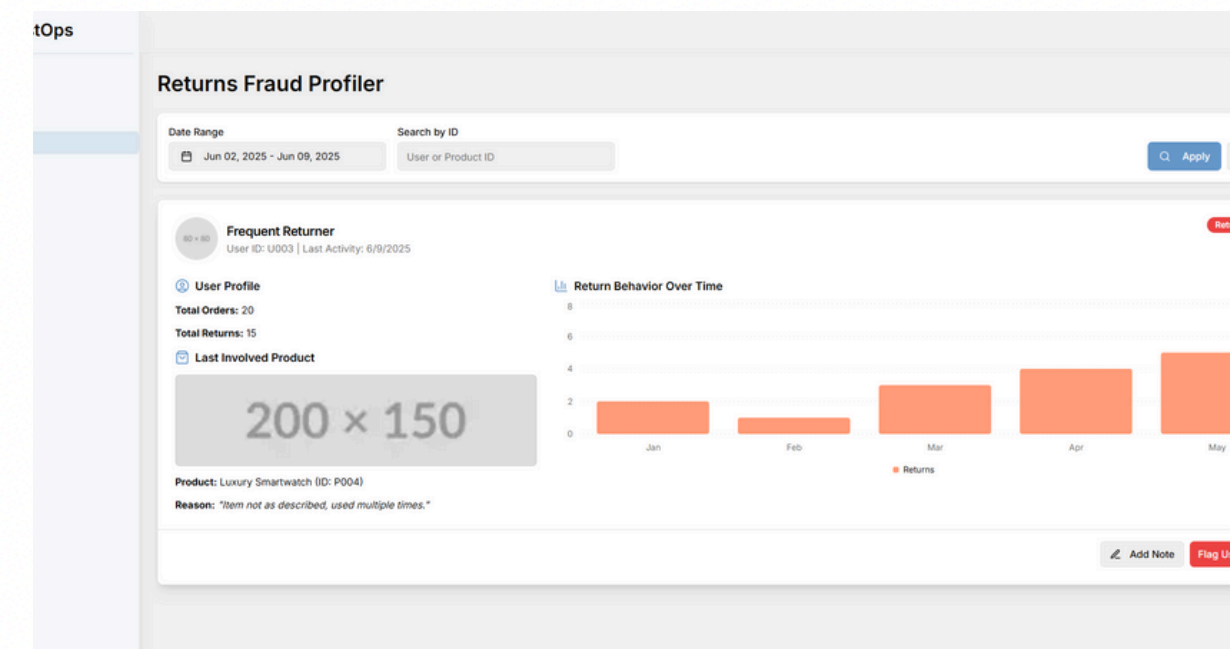
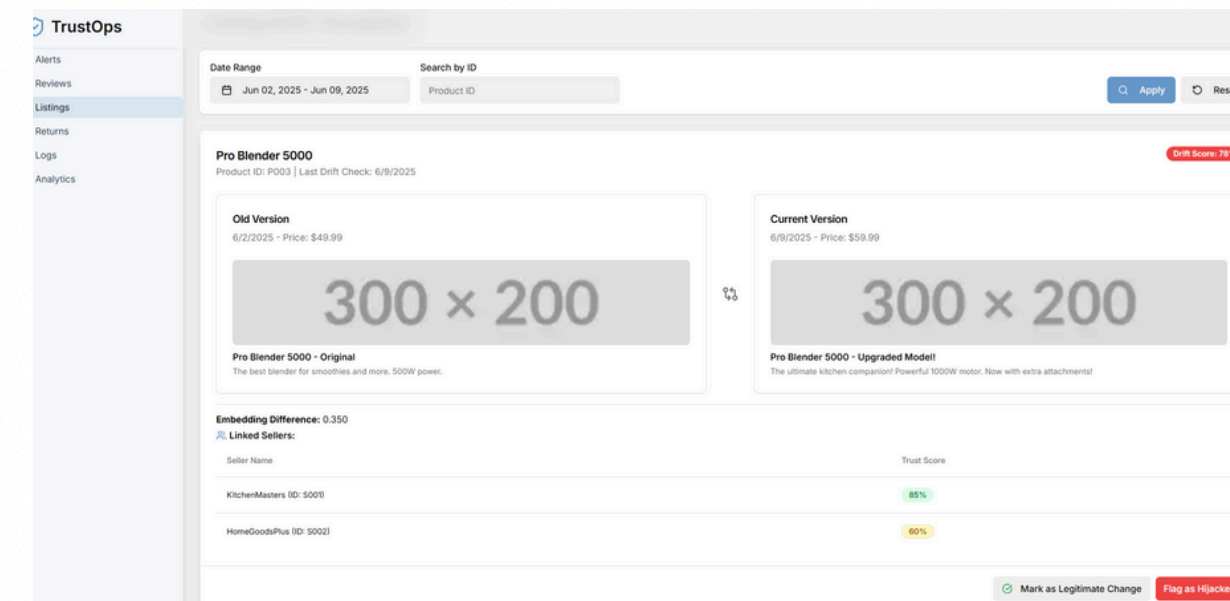
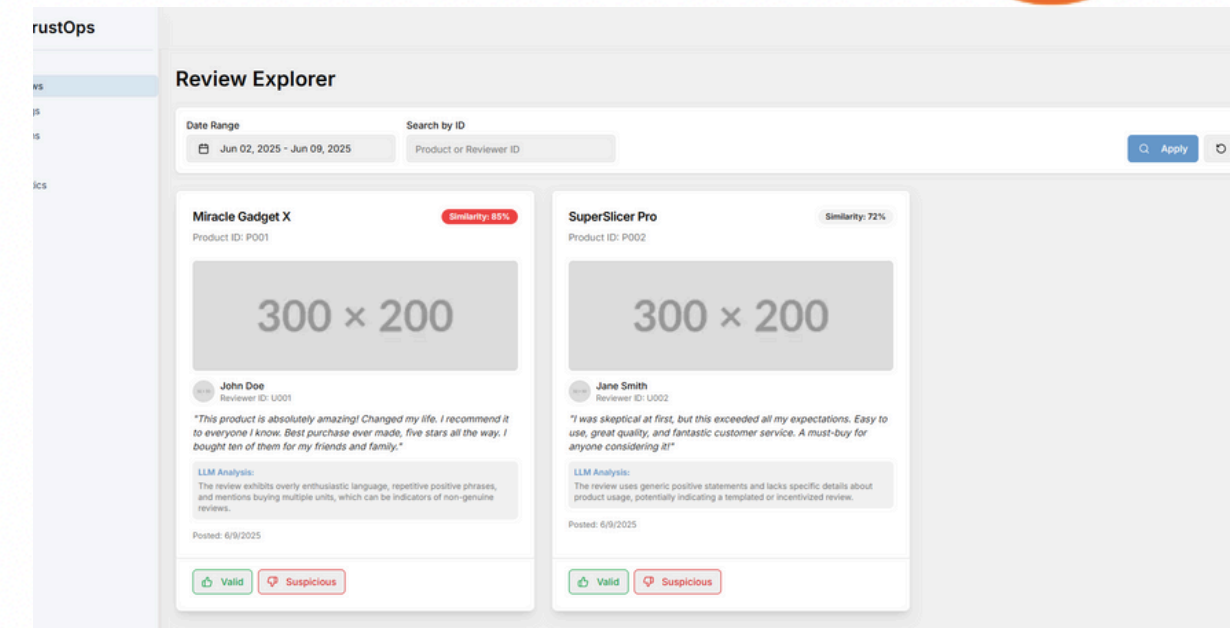
Uses LLM-powered analysis to assess the authenticity of each review based on linguistic patterns, sentiment alignment, and behavioral context. Flags suspicious indicators such as AI-generated text, overly templated content, or reviews that don't match the product's features. Also incorporates reviewer trust scores based on past activity, helping moderators prioritize high-risk signals efficiently.

Drift Visualizer :

Continuously tracks and compares changes in product listing content — such as title, description, and images — to detect semantic drift. This module is key in identifying listing hijacks, where unauthorized sellers subtly or drastically alter trusted listings to redirect traffic or sell counterfeits. Visual timelines highlight when and where the changes occurred, enabling quick intervention by moderators.

Return Profiler :

Surfaces patterns in customer return behavior across time, products, and categories. Identifies return abuse cases such as serial high-value returns, unusual refund frequency, or return loops by the same account. The profiler enables Trust & Safety teams to detect and investigate fraud rings and policy loophole exploiters, reducing losses and improving platform integrity.





CHALLENGES AND KEY LEARNING



Challenges Faced & How We Overcame Them

Top Challenges:

-  Labeling training data for fake reviews and real abuse cases
-  Handling false positives, especially with sarcastic or humorous reviews
-  Real-time inference speed with large models
-  Graph complexity for collusion patterns in massive datasets

What We Learned:

Custom thresholds for review confidence reduce noise

LLM chaining improves multi-step review classification

Graph pruning & prioritization help scale with big data

Moderator feedback loop is crucial for improving trust scores



FUTURE SCOPE AND ROADMAP



What's Next: Future Enhancements

Upcoming Features:

-  Voice-based AI agents for moderation (e.g., handling voice reviews)
-  Blockchain-backed reviewer reputation
-  AI auto-tagging for listing violations (e.g., prohibited content)
-  User-facing transparency tools (like “verified reviewer” trust badges)

Expansion Possibilities:

Fraud detection in Amazon Logistics & Returns

Advertising integrity (fake ad boosts, review-to-ads manipulation)

Integrate with Alexa Reviews, Kindle content moderation, etc.



THANK YOU & Q&A



Let's Build a Safer Marketplace — Together

Our platform is built with Amazon's core principles in mind:

Customer Obsession, Operational Excellence, and Long-Term Trust.

Thank You!

We welcome your feedback and questions.